Response to the Editor and Reviewers

Manuscript: LLM-guided headline rewriting for clickability enhancement without clickbait

We thank the editor and both reviewers for their careful and constructive reports. We have expanded the experimental program, converting a manuscript whose central claims rested on qualitative examples into one in which every claim is backed by quantitative evidence with independent metrics, item-matched significance testing, and confidence intervals. In the revised manuscript, every passage changed in this revision is highlighted in yellow; a white-background HTML mirror and a public data/code/model deposit accompany the submission.

The revised Section 4 (Experiments and Results) now reports, all on held-out data and with independent evaluators wherever possible:

- **external validation of both guide models on entirely human-authored corpora (4.5);**
- *prefix-level validation of the guide on real headlines (4.6, Figure 5);*
- **a quantified Reuters-neutrality check of the source pool (4.7);**
- *a four-cell dual-guidance ablation on 300 held-out headlines (4.8, Table 6, Figure 6);*
- *a comparison against the DExperts and GeDi decoding-time baselines showing that dual control decouples engagement from clickbait (4.9, Table 7, Figure 7, Table 8);*
- *a rater-based evaluation with three independent annotators (4.10, Table 9);*
- *a robustness and factuality analysis (4.11, Table 10);*
- *cross-domain generalization to non-Reuters headlines (4.12, Table 11);*
- *transfer to a second base-model family, Qwen2.5-7B (4.13, Table 12).*

Each reviewer point is answered individually below, with the section and numbers that now address it. Part 2 lists improvements we made that were not explicitly requested.

Part 1 — Point-by-point responses

Reviewer 1

R1-1 — Generalization to naturally occurring clickbait / validation on public real-world datasets. Addressed (Sections 4.5–4.6). Both guides are now evaluated on two entirely human-authored, publicly available corpora never seen in training: Chakraborty et al. 2016 (32,000 headlines) and the Webis Clickbait Corpus 2017 (2,459 graded), each with 95% bootstrap CIs. The clickbait guide reaches AUROC 0.955 on Chakraborty and 0.740 on the noisier graded Webis; the ten attribute dimensions all correlate significantly with human clickbait (Spearman 0.34–0.45, every $p < 1\times10^{-60}$). A new prefix-level test (Section 4.6, Figure 5) shows the guide predicts the human label from an early prefix, with AUROC rising monotonically from 0.891 at 30% of the headline to 0.954 at full length. Every clickbait number in the rewrite evaluation is produced by a separate DistilBERT detector trained only on the human-authored Chakraborty data, never by our own guide.

R1-2 — Central claims supported only by qualitative examples; add quantitative and human evaluation. Addressed (Sections 4.8, 4.10). Over 300 held-out headlines we now report BERTScore-F1, DeBERTa-v3 NLI entailment, independent attribute realization (an LLM judge), independent clickbait probability, and fluency (perplexity) for each configuration, with item-matched significance. We additionally conducted a rater-based evaluation (Section 4.10) with three independent human annotators over 1,500 rewrites, obtaining good-to-excellent reliability (ICC(2,k) 0.82–0.94); annotator faithfulness correlates with automatic BERTScore (Spearman $\rho = 0.61$, $n = 1500$). With item-matched significance the human ratings confirm the two designed effects: the brake lowers perceived clickbait (dual 2.03 vs no-guidance 2.61 on a 1–5 scale, $p = 1\times10^{-7}$) and the dual guidance is rated more faithful than the no-guidance baseline (3.58 vs 2.89, $p = 1\times10^{-7}$) and than the DExperts and GeDi baselines (both $p < 1\times10^{-13}$). On engagement we are explicit about a tradeoff: the human 1–5 rating is lower at the dual operating point than at no-guidance (2.90 vs 3.46), because perceived click-appeal is entangled with clickbait cues. Engagement is nonetheless controllable, not lost: the positive guide raises perceived engagement on its own (engage-only 3.51 vs 3.46), and an independent LLM engagement judge scored across the α–β grid ($n = 840$) holds engagement at or above the no-guidance baseline at the operating point (0.925 vs 0.900) while cutting clickbait, with a lower-brake point (4, 1) higher still — so engagement is a controllable axis set by α (Section 4.10). The full annotation workbooks and per-item ratings are released in the deposit.

R1-3 — Dual-guidance controllability insufficiently validated; add ablations and baselines. Addressed (Sections 4.8–4.9). The dual-guidance mechanism is dissected by a four-cell (α, β) ablation, showing the two guides exert the designed, independent, opposite effects (the engagement guide raises independent tactic realization from 0.48 to 0.51; the brake lowers the independent clickbait score from 0.079 to 0.057). We compare against the two representative decoding-time methods DExperts and GeDi, each with its own strength sweep. The dual guide adds an independent clickbait brake (Table 7, Figure 7) that the single-axis methods lack: a lever that lowers induced clickbait separately from the engagement weight. Because their single control couples engagement and clickbait, the baselines reach the tactic at low clickbait only at low steering; any stronger steering drives their induced clickbait up steeply (DExperts 0.06 to 0.73 across its sweep; GeDi to 0.59) with fidelity falling in step, whereas the brake holds induced clickbait low across the range. For a fair comparison DExperts is decoded with a repetition penalty (1.3) and a headline-length cap, since unconstrained greedy product-of-experts decoding degenerates into repetition at high steering strength.

Reviewer 2

R2-M1 — Unsupported intro claims; ungrounded, unvalidated rubric. Addressed. The opening paragraphs now carry supporting citations, and the rubric is positioned as an extension of the attribute-attribution work of Nofar et al. 2025. The rubric is validated empirically: a blinded relabeling study (Section 4.10, C1) over 150 headlines yields substantial inter-rater agreement (mean Fleiss $\kappa = 0.70$).

R2-M2 — Central claims not supported by quantitative evidence. Addressed. Every claim in the Conclusion is now tied to a measured result in Section 4: the brake measurably lowers both the independent detector score and human-rated clickbait (dual 2.03 vs 2.61, $p = 1\times10^{-7}$), independent human annotators rate the dual guidance more faithful than the no-guidance baseline (3.58 vs 2.89, $p = 1\times10^{-7}$), and against the baselines the dual control holds induced clickbait low at matched tactic realization while the single-axis methods' clickbait rises steeply with steering.

R2-M3 — Circularity of the evaluation loop. Addressed (Section 4.5). Both guides are re-evaluated on entirely human-authored corpora, scoring AUROC 0.955 / 0.740 there alongside the in-distribution synthetic score (0.9998). Every clickbait number in the rewrite evaluation uses an independent DistilBERT detector trained only on human data, which breaks the loop the reviewer identified.

R2-M4 — No baselines or ablations. Addressed (Sections 4.8–4.9, Tables 6, 7, 8; Figures 6, 7). Both are now present: the full four-cell ablation and the DExperts and GeDi baselines, with a decoupling analysis (Table 7, Figure 7) and a qualitative side-by-side (Table 8) showing that when the single-axis baselines are pushed to high steering their induced clickbait rises steeply, whereas the dual guide's independent brake keeps it low. DExperts is decoded with a repetition penalty and a headline-length cap for a fair comparison.

R2-M5 — Missing implementation details / reproducibility. Addressed (Section 3.6). The base model (Meta-Llama-3-8B-Instruct), top-k = 50, the log-domain objective and its swept weights, the tuning grid, cloud hardware, guide-training settings, the full synthetic-data generation procedure (batch size, tactic sampling, JSON self-repair), and per-headline wall-clock are all specified. All prompt templates, hyperparameters, data, code, and the trained guide checkpoints are released in a public deposit.

R2-M6 — Apparent Table 2 inconsistencies; validate tactic-label fidelity. Addressed (Section 4.10, C1). A blinded rubric re-labeling yields $\kappa = 0.70$ and a mean intended-vs-consensus tactic F1 of 0.44, matching the attribute model's own recoverability and reflecting the graded, fine-grained nature of the tactics. The synthetic examples in Table 2 were also refreshed from the authoritative generation run.

R2-M7 — Attribute model may be too weak to guide reliably. Addressed (Sections 4.8, 3.7). The (0,0)→(4,0) contrast isolates the positive guide's contribution directly, and attribute realization is measured by an independent LLM judge rather than the guide itself, so classifier-noise propagation is quantified rather than assumed. The judge's own reliability is cross-checked against the rater study's recoverability (Section 3.7).

R2-M8 — Some controlled outputs contradict the framework's goals. Addressed (Section 4.11, Table 10). Each boundary case is discussed explicitly, and the embellishment concern is quantified: a stricter new-fact judge (validated on faithful-paraphrase controls that score ≈ 0.02 and fact-injected controls that score 1.0, AUROC ≈ 1.0) shows the dual operating point introduces no more genuine new facts than an unguided rewrite (0.20 vs 0.18 for the no-guidance baseline, not significant), so the guidance does not trade faithfulness for engagement. The apparent embellishment is the rhetorical-question form, not fabricated content.

R2 (review question) — "No statistical analysis (single runs, no variance, no significance testing)." Addressed throughout. Every reported comparison uses item-matched Wilcoxon signed-rank tests; all table estimates carry 95% percentile-bootstrap confidence intervals, now including the single-axis tactic values in Table 7. Matched-pairs rank-biserial effect sizes are reported for the key contrasts — the ablation Paired-win column of Table 6 (r = 0.19–0.22) and the rater faithfulness contrasts (dual vs no-guidance / DExperts / GeDi, r = 0.67 / 0.82 / 0.91), collected in Section 3.7. Holm correction is applied within two explicit families (the four ablation contrasts against the (0,0) cell; the cross-method rater faithfulness comparisons), and every reported significant p-value remains significant after correction within its family (Section 3.7).

R2-m1 — Cross-reference errors. Fixed. The methodology subsections referenced in the introduction now exist and resolve correctly.

R2-m2 — Reuters-neutrality assumption unverified. Addressed (Section 4.7). The independent detector was run over the full 20,825-headline source pool: only 0.61% exceed the clickbait threshold, confirming the neutral baseline rather than assuming it. The domain scope is stated explicitly in the manuscript.

R2-m3 — Is the dataset split stratified? Clarified (Sections 4.2–4.3). The split is performed uniformly at random at the source-headline level (not stratified by tactic); because every source contributes both a neutral and a clickbait sample, class balance is preserved by construction.

R2-m4 — Figure y-axis / caption mismatch. Fixed. Figure 3 now labels the axis and caption consistently; the per-tactic figure was regenerated from the current guide.

Part 2 — Additional improvements (not explicitly requested)

Beyond the reviewers' requests, we strengthened the manuscript as follows; all are highlighted in the manuscript.

1. *Full statistical protocol. Bootstrap CIs, effect sizes, and multiple-comparison correction are applied uniformly across Section 4, not only where variance was questioned.*
2. *Cross-domain generalization (Section 4.12, Table 11). The same guides, at the same operating point, reproduce their designed effects on 150 human-authored non-Reuters headlines (a different register): the brake lowers independent clickbait (0.097 → 0.061) and fidelity is preserved (BERTScore 0.22–0.28), extending the demonstrated scope to a second register.*

3. *Transfer to a second base generator (Section 4.13, Table 12). We frame this as a model-family transfer check rather than a second demonstration of the decoupling: the same guides and operating point run on Qwen2.5-7B-Instruct with source fidelity and fluency preserved (BERTScore and perplexity at or better than the Llama-3 configuration). The Table 12 tactic column now reports the independent LLM judge (replacing the guide-based circular score); on Qwen it is flat across cells (≈ 0.50), so the sharp brake-induced drop seen under the circular score was an artifact, not a real loss. Because Qwen's rewrites sit near the detector floor throughout, the decoupling itself is established on the Llama results (Table 7, Figure 7), not re-demonstrated here. For multilingual base generators we add a language-consistency constraint to the decoder (Section 3.6).*

4. *Factuality calibration (Table 10). The new-fact judge and NLI entailment are calibrated on labelled controls, giving the fidelity analysis a verified basis.*

5. *Qualitative depth. A worked "engagement continuum" for one headline across the α sweep (Section 4.4), a full four-cell grid on three headlines (Table 4), and a qualitative cross-method comparison (Table 8) make the control behaviour concrete.*

6. *Objective and evaluators. The combined objective uses the canonical log-domain FUDGE factorization; all evaluators (independent DistilBERT detector, LLM tactic judge, DeBERTa-v3 NLI, distilGPT-2 perplexity) are independent of the guides.*

7. *Complete citation coverage. All datasets, methods, models, and evaluation metrics are now cited, including the base generators (Llama-3, Qwen2.5, GPT-4o) and the evaluation models/metrics (BERT, DistilBERT, DeBERTaV3, BERTScore, GPT-2).*

8. *Reproducibility deposit. Data, code, and the trained guide checkpoints are released publicly, with per-item score files behind every table.*

9. *Presentation. Consistent Scientific Reports reference style, a white-background HTML mirror, and full yellow highlighting of all revisions.*

Each central claim is now backed by the data and independent evaluators described above. We thank the reviewers again for feedback that strengthened the work.

Sincerely, The authors